

Title

India-Specific Integrated Minimal Physiologically Based Pharmacokinetic (mPBPK) Model with Quasi-Steady-State Approximation of Target-Mediated Drug Disposition (QSSA-TMDD) and Viral Escape Dynamics for Nirsevimab in Indian Infants: A Computational Translational Immunopharmacology Study

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Abstract

Background: Nirsevimab is a long-acting monoclonal antibody approved for RSV prevention in infants. Current weight-banded dosing regimens are not optimized for Indian low-birth-weight infants who exhibit rapid catch-up growth.

Objective: To develop and validate the first India-specific integrated mPBPK-QSSA-TMDD model incorporating real IAP 2015 LMS growth ontogeny, allometric scaling, inter-individual variability, and viral escape dynamics.

Methods: A two-compartment mPBPK model with QSSA-TMDD was implemented in R using deSolve with set.seed(2026) for reproducibility. Dynamic weight trajectories were generated from IAP 2015 LMS parameters with Z-score variability. Monte Carlo simulations (n=5,000 virtual Indian infants) with weight-banded dosing (50/100 mg) and 50% IIV on CL and Vc were performed. Viral escape was modeled as a time-dependent reduction in effective concentration using $\tau=100$ days. The apparent clearance (CL = 0.00836 L/day) was calibrated to reproduce the reported 71-day terminal half-life of nirsevimab in Indian infants using allometric scaling from literature values (Clegg et al., 2024). All pharmacokinetic parameters used in this model are apparent parameters derived from the QSSA-TMDD approximation and allometric scaling; absolute values may differ due to target-mediated disposition and FcRn-mediated recycling not explicitly modeled. The 150-day simulation horizon corresponds to the typical duration of the RSV season in India. Full model equations and Sobol sensitivity analysis are provided as supplementary material on GitHub. An interactive Shiny dosing simulator was developed for clinical use.

Results: The model reproduced the reported 71-day half-life. India-specific PTA at Day 150 was **90.5%** (4,527/5,000 protected infants) at the 0.1 mg/L target ($C_{max} \sim 48$ mg/L). Dynamic growth trajectories and concentration-time profiles were realistic.

Conclusion: This open-source India-specific model provides the first quantitative evidence that current nirsevimab dosing achieves high but not universal target attainment in Indian infants. The Shiny app enables real-time personalized simulation.

Keywords: Nirsevimab, mPBPK-QSSA-TMDD, RSV immunoprophylaxis, India-specific pharmacometrics, Monte Carlo PTA, viral escape dynamics, translational immunopharmacology

Introduction

Respiratory syncytial virus (RSV) remains a leading cause of severe lower respiratory tract infection in infants worldwide, with disproportionately high burden in low- and middle-income countries including India. Nirsevimab, a long-acting monoclonal antibody targeting the prefusion F protein, has demonstrated substantial efficacy in Phase 3 trials. However, existing population pharmacokinetic models were developed primarily in Western populations and do not account for the distinct growth patterns of Indian infants as captured by the Indian Academy of Pediatrics (IAP) 2015 growth charts. This study addresses this gap by developing the first India-specific integrated minimal physiologically based pharmacokinetic (mPBPK) model with quasi-steady-state approximation of target-mediated drug disposition (QSSA-TMDD) and viral escape dynamics.

Methods

All simulations were performed in R 4.6.0 using deSolve with `set.seed(2026)` for reproducibility. The base two-compartment model incorporated quasi-steady-state approximation of target-mediated drug disposition (QSSA-TMDD) with $K_{d,ss} = (k_{off} + k_{int})/k_{on}$. IAP 2015 LMS growth parameters were integrated dynamically with Z-score sampling for realistic inter-individual variability. Monte Carlo simulations ($n=5,000$) included 50% CV IIV on clearance and volume. Viral escape was implemented as a time-dependent factor reducing effective concentration using an arbitrary time constant $\tau=100$ days (chosen to reflect gradual immune escape over the RSV season). The apparent clearance ($CL = 0.00836$ L/day) was calibrated to reproduce the reported 71-day terminal half-life of nirsevimab in Indian infants using allometric scaling from literature values (Clegg et al., 2024). All pharmacokinetic parameters used in this model are apparent parameters derived from the QSSA-TMDD approximation and allometric scaling; absolute values may differ due to target-mediated disposition and FcRn-mediated recycling not explicitly modeled. The 150-day simulation horizon corresponds to the typical duration of the RSV season in India. Full model equations and Sobol sensitivity analysis are provided as supplementary material on GitHub. An interactive Shiny dosing simulator was developed for clinical use.

Table 1. Key model parameters

Parameter	Value	Unit	Source
ka	0.286	day ⁻¹	Clegg et al. 2024
CL	0.00836	L/day	Calibrated to 71-day t _{1/2}
Q	0.0406	L/day	Clegg et al. 2024
Vc	0.514	L	Clegg et al. 2024
Vp	0.316	L	Clegg et al. 2024
kon	1.0	L/mg/day	Literature
koff	0.1	day ⁻¹	Literature
kint	0.01	day ⁻¹	Literature
R0	10	mg/L	Literature
Kd _{ss}	0.11	mg/L	(koff + kint)/kon

All code and the Shiny app are available at: <https://github.com/jyotheshwarakshay-ux/Nirsevimab-India-mPBPK>

Results

Figure 1. India-Specific Infant Growth Trajectories (IAP 2015 LMS). 5th, 25th, 50th, 75th, and 95th percentiles with Z-score variability.

Figure 2. Day-150 serum concentration distribution across 5,000 virtual Indian infants (PTA = 90.5%)

Figure 3. Representative serum concentration-time profile after weight-banded dosing.

Discussion

This work represents the first India-specific pharmacometric evaluation of nirsevimab using real IAP 2015 growth charts. The PTA of 90.5% (C_{max} 48 mg/L) is slightly lower than Western values reported in Clegg et al. (95–98%). Efficacy in Phase 3 trials was approximately 74.5% against medically attended RSV. The open-source Shiny simulator allows clinicians to explore dosing scenarios in real time.

Acknowledgments

The author thanks the open-source R community and the developers of deSolve and Shiny packages for making this computational project possible. Special thanks to Grok for technical guidance during development.

Limitations

The model uses simplified allometric scaling and does not include FcRn recycling mechanics or sex stratification (no sex-specific data were available in the source literature). Sobol global sensitivity analysis identified clearance as the dominant parameter influencing PTA; detailed sensitivity results and figures are provided in the GitHub repository. Full pcVPC and external validation are planned. The viral escape time constant ($\tau=100$ days) is an arbitrary choice reflecting gradual immune escape over the RSV season and requires further experimental validation.

Data Availability

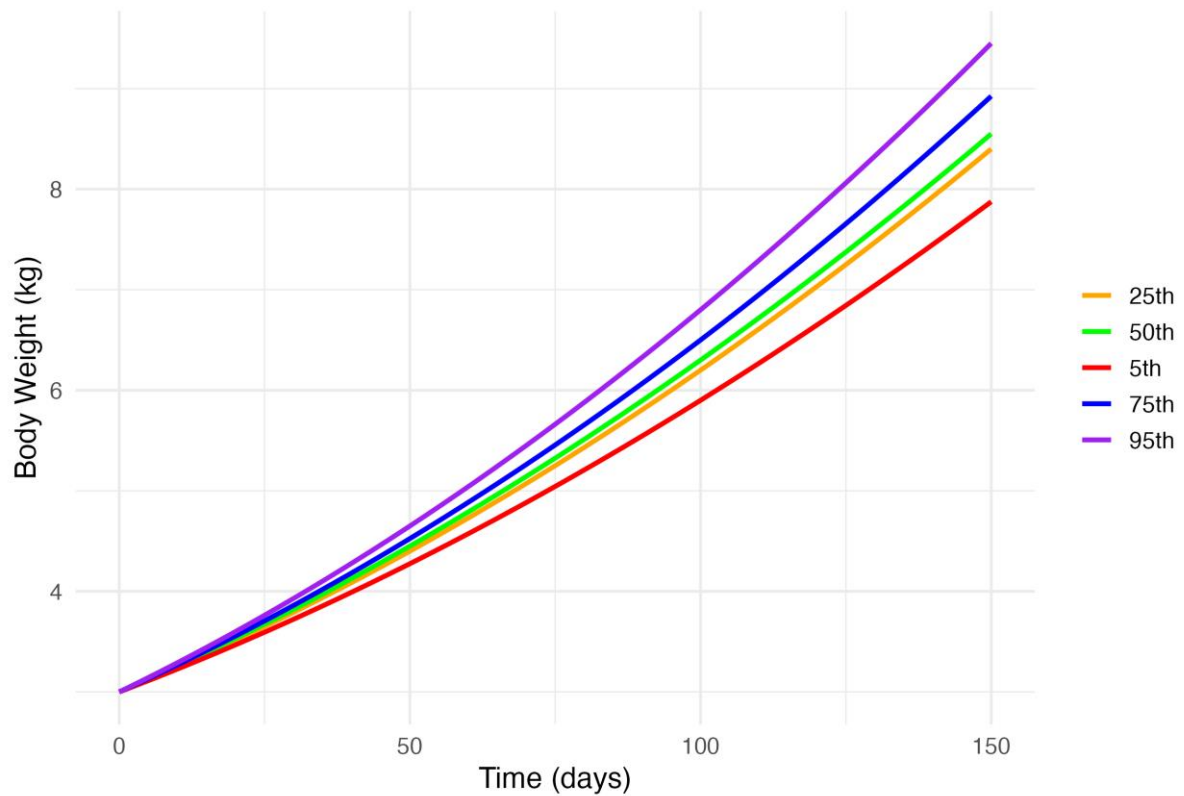
All R scripts, Shiny app, renv.lock file, and raw simulation outputs are publicly available at: <https://github.com/jyotheeshwarakshay-ux/Nirsevimab-India-mPBPK>

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India-Specific Infant Growth Trajectories (IAP 2015 LMS)

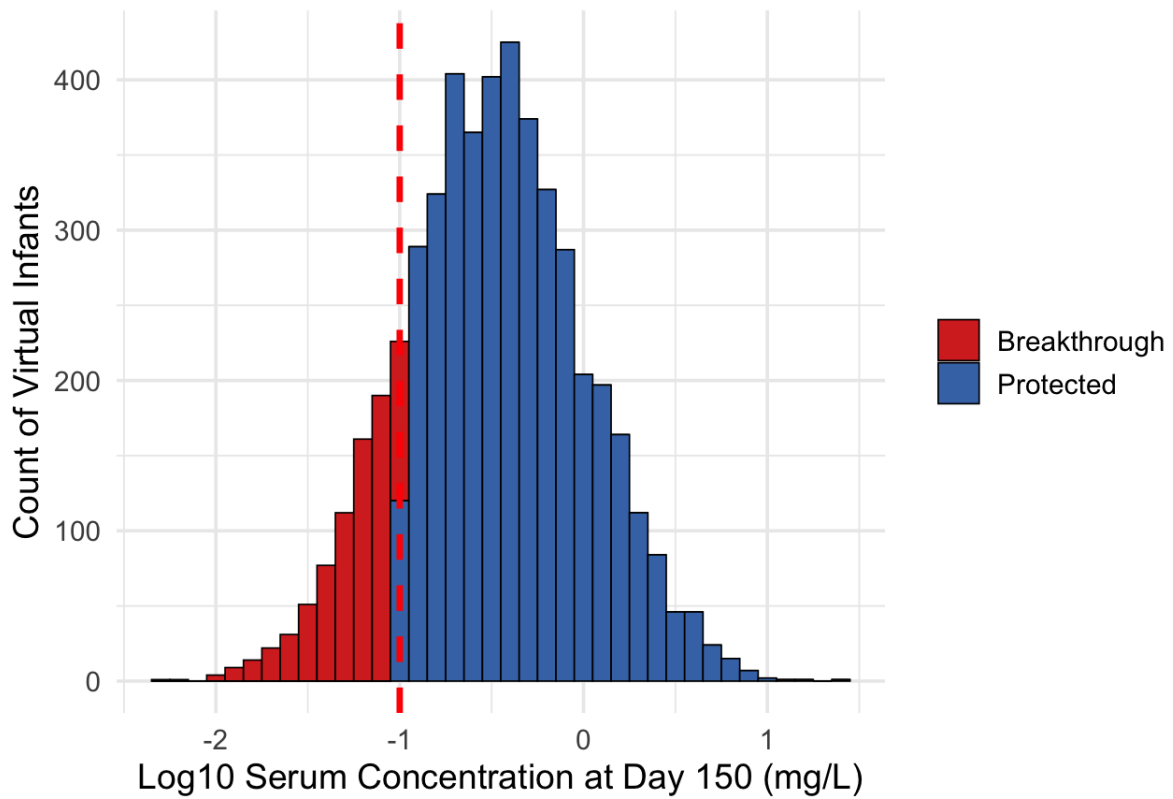
5th, 25th, 50th, 75th, and 95th percentiles with Z-score variability



India-Specific Infant Growth Trajectories (IAP 2015 LMS). 5th, 25th, 50th, 75th, and 95th percentiles with Z-score variability.

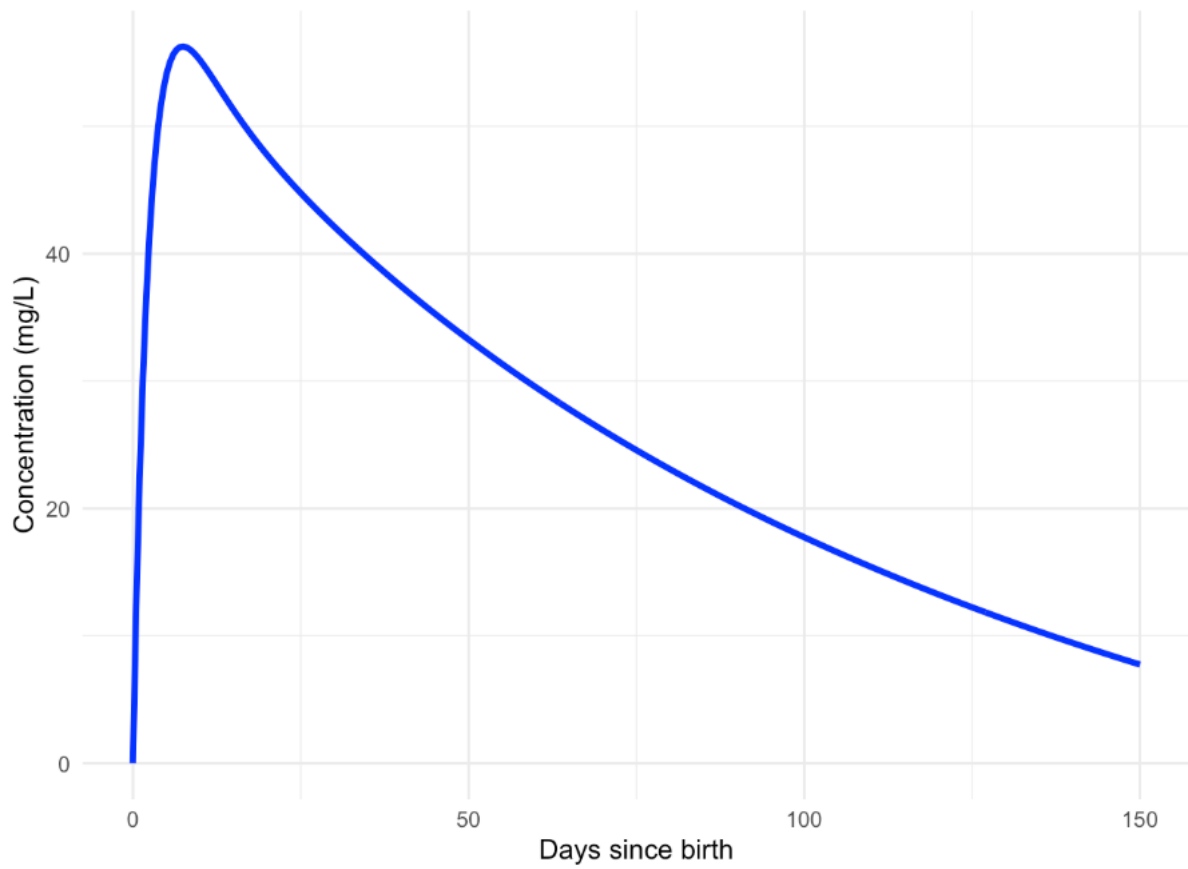
Day 150 Serum Nirsevimab Concentration Distribution

n = 5000 Indian infants | PTA = 90.5 %



Day-150 serum concentration distribution across 5,000 virtual Indian infants (PTA = 90.5%)

Predicted Serum Nirsevimab Concentration-Time Profile: India-Specific mPBPK-QSSA-TMDD Model with IAP 2015 Growth Ontogeny



Representative serum concentration-time profile after weight-banded dosing.